SafeHire: An AI-Powered Service Hiring Platform

Weekly Progress Report 5

Afridur Rahman Khan Mim ID: 2231521642
Nazia Faruque Diya ID: 2222666042
Department of Electrical and Computer Engineering
North South University, Bangladesh
Course: CSE299 Junior Design Project (Section 04)
Group No.: 7

I. Development Progress

During this development phase, significant progress was made in enhancing the SafeHire platform by implementing advanced features related to job matching, transaction management, payment simulation, and user interaction improvements. The system evolved into a complete service workflow system.

A. Job Matching System Implementation

An AI-based matching system was integrated to recommend suitable workers for posted jobs based on skills and experience.

B. Hiring Workflow Development

Employers can now select and assign workers to jobs, updating job status from “Open” to “Assigned.”

C. Transaction System Integration

A structured transaction system was developed to track hiring records and payment details.

D. Payment Simulation Feature

A simulated “Pay Now” system updates transaction status from “Assigned” to “Completed.”

E. Review and Rating System

Employers can rate and review workers after job completion, improving credibility.

F. User Interface Improvements

Modern UI using Tailwind CSS with a consistent dark theme across pages.

G. Data Validation and Error Handling

Improved backend validation and user feedback mechanisms.

H. System Integration and Testing

All modules were integrated and tested to ensure a complete workflow.

I. Backend Development

The backend of the SafeHire platform was developed using Flask and Flask-SQLAlchemy. Core functionalities such as job management, worker assignment, transactions, payment simulation, and review submission were implemented through structured routes. Database models ensured efficient data storage and retrieval.

J. Frontend Development

The frontend was developed using HTML, Tailwind CSS, and JavaScript. A consistent dark theme was applied across all pages. Interactive components such as modals, buttons, and tables were implemented to enhance user experience.

K. Tools and Technologies Used

Flask, Flask-SQLAlchemy, SQLite, Tailwind CSS, JavaScript, GitHub, and Visual Studio Code were used throughout development for backend, frontend, and version control.

II. Conclusion and Future Plan

This development phase successfully transformed the SafeHire platform into a complete and functional service hiring system. The platform now supports a full workflow from job posting to payment and review. The system demonstrates strong architecture, consistent UI, and reliable functionality suitable for final academic evaluation. As the final stage, focus will be on minor refinements, presentation, and documentation. Future improvements may include real payment integration, advanced AI matching, and deployment for real-world usage.